
Leveraging Pre-Trained Transformers for Sentiment Analysis: An Empirical Study on IMDb Movie Reviews

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Abstract

This study evaluates the performance of transformer-based models in sentiment analysis focusing on the IMDb Review Dataset, a widely used benchmark comprising 50,000 polarized movie reviews. While earlier research has employed classical machine learning and deep learning models, recent advances in natural language processing offer the potential for enhanced performance. I fine-tune DistilBERT (a lightweight version of BERT) on the full IMDb dataset and assess its classification accuracy relative to state-of-the-art approaches using the same dataset (notably SVMs, CNNs and LSTMs). Minimal preprocessing is applied to preserve semantic nuance and performance is measured using accuracy, precision, recall, F1-score and confusion matrices. The resulting model achieves a test accuracy of 93.19%, outperforming previous state-of-the-art approaches and illustrating the capability of compressed transformer models to balance efficiency and performance. This analysis suggests that pre-trained transformers, even in distilled form, provide a compelling option for sentiment analysis tasks, especially when balanced against training time and inference efficiency. Important limitations include the absence of cross-validation and external domain testing, both of which offer avenues for future investigation.

1 Introduction

Sentiment analysis is a key component of natural language processing (NLP) that focuses on identifying and classifying the emotional tone expressed in text (Liu, 2022). It aims to detect opinions, emotions or attitudes that are usually categorised as positive, negative or neutral, but recent advances have enabled more nuanced detection of emotions. Sentiment analysis scholarship has seen significant progress in recent years, primarily through the use of machine learning (ML) and deep learning (DL) techniques and more recently with large language models (LLMs) and pre-trained models (Jim et al., 2024). Applications of these techniques span a variety of sectors: in marketing, it informs customer satisfaction and campaign effectiveness; in finance, it assists in analyzing market sentiment; in e-commerce, it guides product improvement; and in healthcare or education, it supports feedback monitoring and service evaluation (Liu, 2022) ; (Jim et al., 2024).

This study explores whether recent NLP advancements—specifically pre-trained models—lead to improved sentiment classification performance compared to earlier models. To do so, I am relying on the IMDB Review Dataset (Maas et al., 2011), a widely used benchmark in sentiment analysis. It consists of 50,000 highly polarized reviews (25,000 positive and 25,000 negative) including a text review and a sentiment label. My approach first reviews benchmark performance across various methods using the IMDb dataset. I then experimented with a more recent technique, DistilBERT, on the same dataset to assess performance improvements.

The remainder of this paper is structured as follows: Section 2. provides a brief literature review. Section 3. outlines my experimental design. Section V presents model implementation and results, while Section 4. provides a conclusion, indicating limitations and future directions.

2 State-of-the-art

This section reviews recent developments in sentiment analysis, beginning with an overview of common algorithmic approaches in sentiment analysis to inform possible choices of methods for our case study (2.1). It then examines models that have been specifically applied to the IMDb dataset, evaluating their design choices and performance outcomes (2.2).

2.1 Recent developments in sentiment analysis

Jim et al. (2024) classify contemporary sentiment analysis approaches into four concentric strata: classical machine learning (ML), deep learning (DL), large language models (LLMs) and multimodal pre-trained models. Classical ML methods—like Support Vector Machines, Naïve Bayes, Random Forest and XGBoost—excel on smaller datasets where features are derived from term frequency or handcrafted cues, offering fast training and high interpretability. As labeled data scales up, DL methods such as Convolutional Neural Network (CNNs) and Long Short-term Memory (LSTMs) outperform traditional ML by capturing semantic patterns and long-range dependencies. Pre-training innovations, including autoencoders and capsule networks, have further boosted accuracy and robustness. The most significant leap comes from transformer-based LLMs like BERT and GPT-3 (Jim et al., 2024), which achieve state-of-the-art results even with minimal supervision and are now standard for fine-tuning and zero-shot tasks. Finally, the frontier lies in multimodal models like DialogueLLM and Wav2Vec 2.0 which integrate text, audio and visual inputs to capture complex sentiments (like sarcasm or emotion in conversations). However, not all of these recent developments in the field of sentiment analysis have yet been applied to the IMDb dataset.

2.2 Sentiment Analysis using IMDb Dataset

2.2.1 Initial Analysis using IMDb Dataset

In their seminal study, Maas et al. (2011)—who introduced the IMDb dataset—proposed a novel method for learning word vectors that incorporate sentiment information. They evaluated it using a newly constructed dataset of 50,000 movie reviews. They trained 50-dimensional word embeddings using both labelled (positive or negative) and unlabelled data. Their method combined two objectives: one unsupervised component that captured semantic relationships between words based on word co-occurrence and one supervised component that adjusted word vectors based on the sentiment of the documents in which they appeared. This way, words appearing in positive reviews would be pushed toward a different region of the vector space than those in negative reviews. To classify entire reviews, they transformed each review into a dense representation using the learned word vectors and then combined this with a standard bag-of-words feature vector. A linear Support Vector Machine was then trained on these combined features to predict sentiment. Their best model, which used both supervised sentiment information and unlabelled data to learn the word vectors, achieved 88.9% accuracy on the test set—outperforming other methods like Latent Semantic Analysis (83.96%) and Latent Dirichlet Allocation (67.42%), as well as traditional bag-of-words models.

2.2.2 Recent Analysis using IMDb Dataset

More recently, Tripathi et al. (2020) extended the work of Maas et al. (2011) by evaluating four classic machine learning classifiers —Naïve Bayes, Logistic Regression, Decision Tree and Random Forest—on the IMDb dataset. They employed two common text vectorisation methods: Count Vectorisation and TF-IDF. With Count Vectorization, the Random Forest classifier achieved an accuracy of 85.84%, while Logistic Regression slightly outperformed it with 87.28% accuracy. When using TF-IDF, performance improved for most models, with Random Forest achieving 85.62% accuracy, though Logistic Regression again led with 89.14%.

Other scholars have applied deep learning methods to the IMDb dataset. Ali et al. (2019) compared performance on the IMDb dataset of four deep learning models : MLP, CNN, LSTM and a hybrid

CNN-LSTM. Their hybrid model, which integrated convolutional feature extraction with sequential LSTM modeling, reached an accuracy of 89.2%, outperforming both standalone CNN (87.7%) and LSTM (86.6%) models and improving upon traditional methods like SVM and Naïve Bayes.

Yenter and Verma (2017) also proposed a CNN-LSTM model with multiple convolutional branches and kernel sizes inspired by the Inception architecture. Their best-performing variant reached 89.5% accuracy on the IMDB dataset, improving over prior CNN-LSTM models by fine-tuning convolutional kernel sizes, dropout rates and LSTM units while addressing overfitting through batch normalization and regularization. Similarly, (Alaparthi and Mishra, 2021) (2019) compared CNN, LSTM and LSTM-CNN models using custom word embeddings. Their results showed CNN achieving the highest F1-score of 91%, surpassing the hybrid and LSTM-only models and highlighting CNN’s efficiency in capturing local semantic features crucial for binary sentiment classification. Finally, Shivaji and Manit (2021) comparing four sentiment analysis approaches on the IMDB dataset highlights BERT’s ability to capture contextual nuances in sentiment classification, achieving an accuracy score of 92.31% compared to 89.4% for logistic regression, 86.8% for LSTM and 63.08% for SentiWordNet.

Collectively, these studies illustrate the increasing prominence of CNN-based hybrid architectures in sentiment analysis using the IMDB dataset, even though the gain in accuracy remains relatively modest compared to the foundational work of Maas et al. (2011).

3 Proposed experiment: a pre-trained model on IMDB reviews

3.1 Pre-trained Model : DistilBERT

While a wide range of machine learning and deep learning recent developments have been applied to the IMDB dataset, there remains a notable gap in the application of pre-trained transformer and LLMs models—an issue highlighted by Shivaji and Manit (2021). To address this, I choose to fine-tune DistilBERT-base-uncased (and not BERT or other LLM because of computation power). DistilBERT is indeed a distilled version of BERT with 6 layers, 12 attention heads and 66 million parameters. It retains over 95% of BERT’s performance while reducing inference time and memory usage by approximately 40% (Sanh et al., 2019). DistilBERT’s bidirectional self-attention over a 512-token window enables it to capture long-range dependencies, polarity inversions and contextual nuances that often challenge recurrent or convolutional models—features cited as key to the success of large language models (Jim et al., 2024).

The model was fine-tuned on the full IMDB dataset (25,000 training and 25,000 testing samples), using a batch size of 16, a learning rate of $2e-5$ and the AdamW optimizer. Training was conducted over three epochs (to limit computing time) with a linear learning rate scheduler and a 10% warm-up phase to stabilize early optimization. Tokenization was handled via the *distilbert-base-uncased* tokenizer with truncation and padding to 512 tokens. These settings reflect current best practices for fine-tuning transformers on medium-scale classification tasks (Sanh et al., 2019).

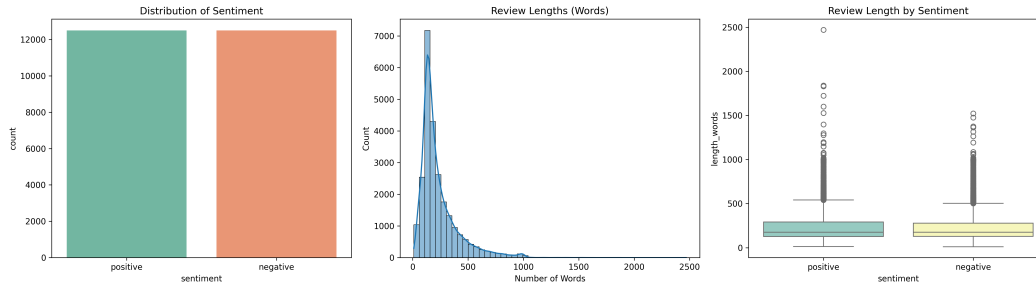
Following the precedent set by Maas et al. (2011), I use the original train/test split without cross-validation to ensure comparability with prior benchmarks and leveraging the dataset’s balanced and curated structure.

The model was trained on a single NVIDIA Tesla T4 (16 GB) using a jupyter notebook and evaluated against top-performing SVM, CNN or LSTM baselines.

3.2 Dataset description and explanatory analysis

The IMDB Review Dataset is a benchmark corpus for sentiment analysis introduced by Maas et al. (2011) and still widely used in recent research (Yenter and Verma, 2017; (Ali et al., 2019); (Tripathi et al., 2020)).

The collection contains 50 000 movie reviews (superior to 30 per title) drawn from the Internet Movie Database and expressly curated for binary sentiment classification. The polarity label is derived from the original IMDB rating. Scores inferior to 4/10 are mapped to negative and scores superior to 7/10 to positive, while middling ratings (5–6) are discarded to avoid ambiguity. The corpus is split 50/50 into 25 000 training and 25 000 test documents, each half preserving a perfect class balance: 12 500 positive and 12 500 negative reviews (Figure 1a). Each data point therefore comprises : 1) a review field with the full, raw HTML, stripped text; 2) a sentiment field (positive or negative).



A first exploratory analysis (Figures 1b and 1c) shows that review length is a bit skewed: the modal document runs to around 120 words yet a long tail stretches beyond 1 000 words, but not necessitating trimming rules. A side-by-side box-plot (Figure 1c) confirms that positive and negative texts share almost identical length distributions; hence document size alone is unlikely to drive classification performance.

As part of our exploratory data analysis, a bag-of-words visualization (Figure 2) was generated to identify frequent lexical patterns across sentiment classes. Both positive and negative reviews center on similar narrative elements (character, story, scene), even if some polarity is observed through evaluative adjectives (as great, love, awful or nothing). This insight can be valuable before applying a model like BERT as it highlights which lexical cues may drive classification (with no clear patterns in our case).



Figure 2: Word Clouds on IMDB dataset

3.3 Data preprocessing

Modern transformer-based language models such as DistilBERT are designed to operate directly on raw text and incorporate robust tokenization strategies (lowercasing, subword segmentation and punctuation handling) thereby obviating the need for extensive manual cleaning (Devlin et al., 2019). In practice, one need only strip residual HTML tags (e.g., using a simple parser) before tokenization; the uncased DistilBERT tokenizer will automatically normalize case, decompose contractions and encode rare or compound words via WordPiece embeddings. Additional preprocessing steps such as accent removal, lemmatization, stop-word filtering, or sentence segmentation are not only redundant but may inadvertently discard linguistic nuances that the model exploits (but this approach is particularly appropriate for English-language corpora and pre-trained uncased models as additional preprocessing may still be required for noisy or multilingual data). Consequently, minimal preprocessing enhances both efficiency and performance when fine-tuning DistilBERT for sentiment classification.

3.4 Metrics

I evaluate my binary sentiment classifier using a suite of complementary performance metrics. Accuracy measures the overall proportion of correct predictions and is particularly interpretable given the perfectly balanced test set. To assess error types, I compute precision (the fraction of predicted positives that are true positives) and recall (the fraction of actual positives that are correctly identified), which together show the model’s propensity for false alarms versus misses (Hossin and Sulaiman, 2015). The F1-score, the harmonic mean of precision and recall, provides a single summary metric that balances these two dimensions (Sokolova and Lapalme, 2009). Finally, the confusion matrix tabulates true positives, false positives, true negatives and false negatives, enabling detailed inspection of misclassification patterns and guiding targeted model improvements

4 Analysis of results

The results demonstrate that the DistilBERT model trained on the full IMDB dataset achieved strong and balanced performance across both sentiment classes (Table 1). The training loss steadily decreased over the epochs: from 0.2912 to 0.065 (Figure 3) indicating effective learning and convergence. On the test set, the model reached an accuracy of 93.19%, with nearly identical precision and recall for both the negative and positive classes (F1-scores of 0.9316 and 0.9322, respectively). The confusion matrix (Table 2) shows a relatively low number of misclassifications, with slightly more false positives (911) than false negatives (791). Further research could include qualitative analysis of misclassified reviews to uncover systematic model blind spots but overall the classifier appears robust, well-calibrated and generalizes well to unseen data.

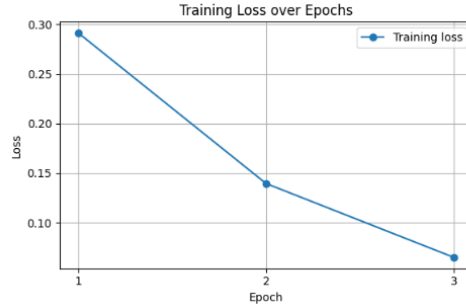


Figure 3: Graphical Representation of loss during training phase

Table 1: Classification Report

Class	Precision	Recall	F1-score	Support
negative	0.9361	0.9271	0.9316	12500
positive	0.9278	0.9367	0.9322	12500
Accuracy		0.9319		25000
Macro avg	0.9320	0.9319	0.9319	25000
Weighted avg	0.9320	0.9319	0.9319	25000

Table 2: Confusion Matrix

		Predicted	
		negative	positive
Actual	negative	11589	911
	positive	791	11709

Table 3: Comparison of model performance on the IMDB dataset

Reference	Dataset	Model	Accuracy (%)
Maas <i>et al.</i> (2011)	IMDb	SVM + sentiment-informed word vectors	88.9
Maas <i>et al.</i> (2011)	IMDb	Latent Semantic Analysis (LSA)	83.96
Maas <i>et al.</i> (2011)	IMDb	Latent Dirichlet Allocation (LDA)	67.42
Yenter & Verma (2017)	IMDb	Inception-style CNN–LSTM	89.5
Ali <i>et al.</i> (2019)	IMDb	CNN	87.7
Ali <i>et al.</i> (2019)	IMDb	LSTM	86.6
Ali <i>et al.</i> (2019)	IMDb	Hybrid CNN–LSTM	89.2
Haque <i>et al.</i> (2019)	IMDb	CNN	90
Haque <i>et al.</i> (2019)	IMDb	LSTM	88
Haque <i>et al.</i> (2019)	IMDb	LSTM–CNN	89
Tripathi <i>et al.</i> (2020)	IMDb	Random Forest (Count Vectorization)	85.84
Tripathi <i>et al.</i> (2020)	IMDb	Logistic Regression (Count Vectorization)	87.28
Tripathi <i>et al.</i> (2020)	IMDb	Random Forest (TF–IDF)	85.62
Tripathi <i>et al.</i> (2020)	IMDb	Logistic Regression (TF–IDF)	89.14
Shivaji & Mishra (2021)	IMDb	SentiWordNet	63.08
Shivaji & Mishra (2021)	IMDb	Logistic Regression	89.41
Shivaji & Mishra (2021)	IMDb	LSTM	86.75
Shivaji & Mishra (2021)	IMDb	BERT	92.31
Guillaume-Le Gall (2025)	IMDb	DistilBERT	93.19

5 Conclusion

Before delving deeper into the implications of these results, it is important to recall their limitations : this study is limited by its exclusive reliance on the IMDb dataset using a fixed train/test split without cross-validation or external evaluation on other sentiment-laden corpora and missing a qualitative analysis of misclassified reviews. Despite these limitations, the DistilBERT model implemented in this analysis achieves state-of-the-art performance on the IMDb sentiment classification task (cf summary Table 4), with an accuracy of 93.19%, slightly surpassing BERT (92.31%) and outperforming earlier deep learning models such as CNN–LSTM hybrids (around 89%).

Given the size of the test set (25,000 review), this gain of 1% accuracy translates to approximately 250 additional correct predictions which is an important margin in domains where classification errors can have important consequences such as health (cf. Tim et al., 2024). Future work should explore cross-domain transferability to assess the model’s generalization to other sentiment-laden corpora such as Amazon, Yelp, or Twitter.

Nonetheless, this improvement comes at a cost. Training DistilBERT on the full IMDb dataset takes approximately 1h40 and requires a GPU. The computational and environmental burden of transformer-based models is substantial and increasingly recognized for its rebound effects (Anthony et al., 2020); (Luccioni et al., 2022). Thus, a broader reflection is needed on the societal trade-offs between marginal performance gains, scientific and societal progress and environmental sustainability.

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